Lab Task 1:

Translate the linear search algorithm into a program which either find the location LOC where ITEM appears in ARRAY or return LOC=0

Code:

```
#include <iostream>
using namespace std;

int main()
{
	const int arrSize = 10;
	int arr[arrSize] = { 21, 99, 150, 212, 320, 420, 510, 603, 834 , 911 };
	int item = 603;
	int loc = 0;

	for (int i = 0; i < arrSize; i++) {
		if (arr[i] == item) {
			loc = i + 1;
			break;
		}
	}

	if (loc != 0) {
		cout << "Item " << item << " found at location " << loc << endl;
	}
	else {
		cout << "Item not found in the array." << endl;
	}
}
```

Output:

```
Item 603 found at location 8
```